

ENERGY STAR HOMES IN MICHIGAN

In 2005, Habitat for Humanity of Detroit built 30 homes as part of the Jimmy Carter Work Project. All 30 homes were certified as Energy Star compliant.

by JACOB CORVIDAE

Each year since 1984, former U.S. President and Nobel Peace Prize Laureate Jimmy Carter, along with his wife, Rosalynn, has led the Jimmy Carter Work Project. The project invites volunteers from around the world to travel to different locations to build Habitat Homes with families who need decent, affordable shelter. My employer, the WARM Training Center, promotes the development of resource-efficient, affordable, healthy homes and communities through education, training, and technical assistance. WARM has been based in Detroit for 25 years and is a Michigan Energy Demonstration Center.

In 2005, Habitat for Humanity of Detroit built 30 homes as part of the Jimmy Carter Work Project for that year. WARM Training Center consulted with Habitat for Humanity of Detroit to review its home designs and train the construction crew so that the homes could achieve Energy Star standards. After the homes were completed, WARM Training Center conducted an independent assessment of the homes to verify that they met the Energy Star standards. All 30 homes were certified as Energy Star compliant.

In 2006 the WARM Training Center received a grant from the Michigan Energy Office to conduct a study of the 30 homes and of their new occupants. WARM staff met with the new homeowners to answer any questions they had about their new home's energy use, and to tell them how they could lower their energy bills.



The Jimmy Carter Center Work Project invites volunteers from around the world to travel to different locations to build Habitat Homes with families who need decent, affordable homes. (Former President Jimmy Carter and Rosalynn Carter are at center left in the photo.)

We gathered billing histories from DTE Energy for the 30 households. Two households were removed from the study because 12 months of data were not available. The study used 12 months of gas (measured in cent cubic feet, or CCF) and electricity (kWh) usage data from September 2005 through August 2006. The usage data for each fuel type were added together to arrive at an annual usage amount for each household. The annual usage amounts for all 28 households were added together, and the total was divided by 28 to find an annual average usage amount for the Habitat Homes. We divided the annual

usage amount by 12 to create monthly average usage amounts.

Since electricity use varies significantly with the number of people living in the household, the homeowners were asked to provide this number. Only 13 of them did so. For these 13 households, we divided the monthly average of kWh usage by the number of people in residence to create the per capita electric monthly average.

For Comparison

Our study did not allow for a formal control group. Nonetheless, we needed

comparison groups in order to understand the data. Three sets of comparison data were gathered:

Standard new-home construction in Michigan. Standard new-home construction data use kilo Btu (kBtu) as a combined measure of kWh and CCF usage. These data were gathered from *Life Cycle Analysis of a Residential Home in Michigan*, a 1998 study by the University of Michigan's Center for Sustainable Systems. The data are based on standard construction of a new 2,450 ft² house in Michigan. The Habitat Homes averaged approximately 2,400 ft².

DTE Energy territory. DTE Energy was able to provide monthly usage amounts for all Michigan homes in the DTE territory. These data were provided for CCF and kWh for 2004 and kWh for 2005. The 2004 data were used for this study. Since weather varies from year to year, the DTE territory homes' heating and cooling needs in 2004 may differ from the homes' heating and cooling needs in the 12-month period assessed for the Habitat Homes.

MES! WARM Training Center conducts a program called Michigan Energy Smart! (MES!). This program is designed specifically for low-income clients who have applied for fuel assistance. WARM took a random sample of 30 MES! clients and gathered billing history information from DTE. This provides a comparison to low-income customers in Detroit, although the comparison is based on customers who have applied for fuel assistance.

We considered that the data for standard new-home construction in Michigan provided the most solid baseline for comparison. Since the weather from the year of collected data on the Habitat Homes affects the comparison, heating degree-days (HDD) for that year were compared to the average annual HDD for Detroit. Data for the Habitat Homes were collected from September 2005 through August 2006. The data for the HDD that this study used were collected from July 2005 through June 2006. While the dates do not match exactly, Detroit typically has only 12 HDD in July and August (out of 6,417 HDD for

the year), which are the only months out of sync between the HDD and billing history data sets.

The kBtu/ft² per year data were divided by the HDD to create an annual Btu/ft²/HDD figure. This figure was compared for the Habitat Homes and standard new-home construction in Michigan. The kBtu combines gas and electric energy use, though gas-only energy use per HDD would be better. Unfortunately, we didn't have data for gas-only energy use in standard new

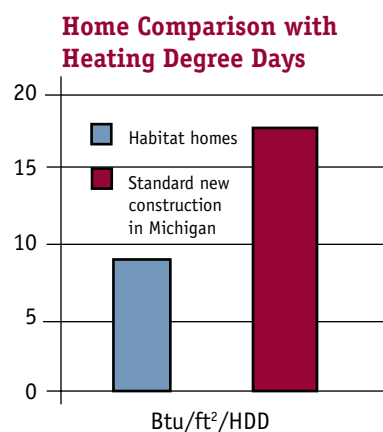


Figure 1. The Habitat Homes built through the Jimmy Carter Work Project used 48% of the energy used by standard new-home construction in Michigan.

construction; we had these data only for the DTE territory—and standard new construction was a better baseline overall. Gas accounts for more than half of the cost of energy, and we incorporated the HDD to come up with a more conservative figure, since we knew that the year these data came from was warmer than usual.

Average Energy Use

The Habitat Homes, on average, used 814 CCF of gas annually. The monthly average CCF use for the Habitat Homes was 68 CCF. This is 41% of the annual CCF use through the DTE territory and 45% of the monthly CCF use of the MES! group. Note that these data are not controlled for variations in temperature and climate for the time period

studied, nor are they controlled for floor area. The Habitat Homes, on average, used 7,100 kWh of electricity annually. The monthly average kWh use for the Habitat Homes was 592 kWh. This is 92% of the annual kWh use through the DTE territory and 98% of the monthly kWh use of the MES! group. Most Habitat Homes were built without A/C, though a few homeowners put in their own units later.

The Habitat Homes, on average, used 110,000 kBtu total energy annually, or 47 kBtu/ft² per year. By comparison, standard new-home construction in Michigan uses 114 kBtu/ft² per year. The Habitat Homes used 41% of the kBtu/ft² per year used by standard new-home construction in Michigan. To further compare, the single lowest kBtu/ft² per year use of a Habitat Home was 34 kBtu/ft² per year, or 30% of the energy used by standard new-home construction in Michigan. The single highest kBtu/ft² per year use of a Habitat Home was 73 kBtu/ft² per year, or 64% of the energy used by standard new-home construction in Michigan. (There was no range for the standard-home baseline—it was derived from a single home studied by the University of Michigan, as we explained above.)

In order to account for differences in weather norms between the Habitat Homes' actual use and the expected use for standard new-home construction in Michigan, we compared Btu/ft²/HDD. The Habitat Homes used 48% of the energy used by standard new-home construction in Michigan when compared by this variable (see Figure 1).

Energy Use per Capita

Electricity use is commonly considered to be correlated with the number of people in a household. This study attempted to examine the correlation between number of residents and use of electricity and gas. Information on the number of residents in each home was gathered for only 13 homes. Given such a small sample, this portion of the data serves only to suggest possibilities for further exploration.

Average monthly electric use for this sample of 13 households was 603 kWh, and average monthly per capita electric use was 176 kWh. Average monthly gas use for this sample was 68.5 CCF, and average monthly per capita gas use was 20.4 CCF.

Why Such Good Results?

While national Energy Star figures show that Energy Star homes typically use 30% less energy than standard-built homes, we found savings of 52% over standard new construction in Michigan. The improvement in energy use is dramatic. Given standard 2006 prices in Michigan, this translates into a savings of at least \$1,500 a year (with rates of \$1/CCF and \$0.10/kWh).

These savings are important for any homeowner, but they may be particularly significant for the affordable housing field and for lenders, since lenders must closely examine homeowner finances before approving a mortgage.

Two reasons most likely account for the dramatic improvement. First, the Michigan Residential Code is less stringent than the 1992 Model Energy Code (MEC), and therefore lags behind most of the country's energy performance standards. Since Michigan starts with a lower standard, bringing a home up to Energy Star standards results in even greater savings. Second, the owners of the study homes all received education on how to keep their bills low.

Training occupants is an important step in maximizing energy savings. Note also that the improvement in total energy use was based primarily on the improvement in the use of gas for heating. This might be expected, given that the Energy Star home standards focus primarily on improvements to the house shell that can reduce heating and cooling energy use. Since heating accounts for most energy use in Michigan's climate, it is not surprising that the Energy Star homes showed the greatest improvement in gas use, when compared to standard construction. The range of total energy used by the Habitat Homes was quite wide; it represented 30%–64% of the

total energy used by the standard new construction in Michigan home. Even the home that used the most energy realized more than the 30% energy savings that Energy Star homes nationally are expected to achieve.

There are probably many reasons for the wide range of energy use in the Habitat Homes. Since the house shells

homes, while electricity—not gas—use is usually considered to be the only energy use that is closely correlated with number of occupants. However, our data showed a similar correlation between number of residents and use of both gas and electricity. Note that while energy use for a household tended to increase with the number of residents,



These Habitat Homes use approximately half of the energy of typical similarly sized homes in Michigan.

were all in similar condition, the main factors that could affect energy use are the appliances and the occupants. While the homes were furnished with Energy Star lighting and new refrigerators, homeowners supplied the other appliances. The type and number of additional appliances account for some of the difference in energy use. Yet, as we have seen, gas use for heating had a far larger effect on overall energy use than did the use of electricity. This suggests that occupant behavior accounted for much of the range in energy use, since the occupants set the temperatures for heating.

The number of occupants seemed to affect total energy use in the Habitat Homes. These results were not entirely expected, given that gas use accounted for most of the energy used in these

the amount of energy used per person decreased. Nonetheless, the per capita data from this study are merely suggestive and should be further explored for insight into how to lessen energy use in homes.

Finally, note that the size of a home clearly affects its energy use. We used kBtu/ft² per year to compare the Habitat Homes with standard new construction in Michigan. This eliminates using the size of a home as a factor in comparing energy use. The fact remains that smaller homes generally use less energy. Yet the Residential Buildings fact sheet from the Center for Sustainable Systems shows that the average size of single-family homes in the United States has increased steadily over the past 50 years. Counteracting this trend is another im-

portant strategy for reducing the energy use of new homes.

Michigan, and the World, Benefit

The WARM Training Center study of Habitat Homes shows the notable utility savings possible for homes built to Energy Star standards in Michigan. It suggests that Energy Star homes built in Michigan will save even more than the 30% energy savings cited in most Energy Star literature, and that these homes may achieve over 50% savings. (The 30% savings is based on national standards for standard-built homes. Since Michigan's baseline is worse, the percentage improvement for achieving Energy Star is better.) Furthermore, this study suggests that homeowner education should help to reduce the residents' energy bills.

With an approximate savings of at least \$1,500 a year at current utility prices (estimated at \$1/CCF and \$0.10/kWh), the improved energy standards of these homes have an important impact on the homeowner's operating costs. Over ten years, the owner of an Energy Star home will save \$15,000 at current utility prices.



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WARM Training Center would like to thank the following groups for their assistance in gathering information for this article: Habitat for Humanity of Detroit staff; the owners of the Jimmy Carter Work Project homes; DTE Energy; and the Michigan Energy Office, which is part of the state's Department of Labor and Economic Growth.

For more information:

Find out more about the WARM Training Center at www.warmtraining.org.

The report, *Life Cycle Analysis of a Residential Home in Michigan*, is available through the University of Michigan for download on the Web at <http://css.snre.umich.edu/main.php>.

The Residential Buildings fact sheet from the Center for Sustainable Systems at the University of Michigan is available for download at http://css.snre.umich.edu/css_doc/CSS01-08.pdf.